Best reallocation of the 2027 US MOU money for deaths, 15 countries (no model changes)

Generated by `audit/reallocation/run_optimal_realloc_2027.py`. Dollar for dollar: US$1 of MOU moved = US$1 of US direct aid in the country, spent the way that country's US direct aid is spent, run through the model itself (America First MOUs, US bilateral schedule only, dashboard defaults; deaths ranges from 400 paired draws).

Bounds. Total unchanged; a receiving country can at most get back its 2021-23 US direct aid (0% cut in 2027: the model has no coverage above baseline, so it is not extrapolated); a giving country can at most give the US aid it still has; each country within +/-10%, 20% or 30% of its signed 2027 amount; one version also limits the total moved to the team's Higher Ceiling (US$133.6M).

Method. Each country's deaths 2026-2030 were computed with the model on a US$1M grid of aid changes. Starting from the Original MOU, US$1M was moved at a time between whichever two countries lowered total deaths the most, in either direction, until no move helped. The model's curves are not straight lines (restoring ART money brings back people cut in 2026; deeper cuts cost fewer deaths per dollar), so this is the best allocation found, not a proven global optimum. Amounts are in whole US$M.

1. Results vs the Original MOU

Positive = better than the Original MOU.

Scenario	Moved US$M	Deaths averted 2026-30	Range (paired)	Deaths averted 2031-35	HIV deaths averted	HIV infections averted	People keeping a service 2027 (six services)	ART patients 2027	Check: aid change US$M (should be 0)
Best within +/-10%	103.0	+11,571	+7,863 to +17,409	+3,705	+7,853	+21,622	+1,334,426	+30,031	+0.00
Best within +/-20%	165.0	+22,200	+15,487 to +36,289	+8,671	+19,228	+54,869	+1,010,586	+95,133	-0.00
Best within +/-30%	208.0	+27,379	+19,312 to +44,668	+10,583	+23,552	+65,964	+2,119,306	+83,363	-0.00
Best, US$133.6M moved (as Higher Ceiling), +/-30%	133.0	+17,984	+12,155 to +28,928	+6,588	+14,689	+35,819	+1,159,285	+76,734	+0.00

Original MOU: 189,629 deaths 2026-2030, 6,914,369 people losing a service in 2027, 455,448 losing ART.

2. The allocations (2027 MOU amounts, US$M; change vs Original in brackets)

Country	Original MOU	Best within +/-10%	Best within +/-20%	Best within +/-30%	Best, US$133.6M moved (as Higher Ceiling), +/-30%	Room to 0% cut US$M	Deaths averted by first +US$1M
Botswana	37.8	40.8 (+3.0)	42.8 (+5.0)	42.8 (+5.0)	42.8 (+5.0)	5.3	154
Burundi	33.1	30.1 (-3.0)	27.1 (-6.0)	24.1 (-9.0)	24.1 (-9.0)	6.9	58
Cameroon	91.5	100.5 (+9.0)	107.5 (+16.0)	107.5 (+16.0)	107.5 (+16.0)	16.5	133
Eswatini	47.0	51.0 (+4.0)	56.0 (+9.0)	57.0 (+10.0)	57.0 (+10.0)	10.9	211
Ethiopia	247.0	254.0 (+7.0)	254.0 (+7.0)	254.0 (+7.0)	254.0 (+7.0)	7.5	155
Kenya	388.3	350.3 (-38.0)	311.3 (-77.0)	272.3 (-116.0)	388.3	11.8	89
Lesotho	55.0	50.0 (-5.0)	66.0 (+11.0)	71.0 (+16.0)	55.0	20.8	116
Liberia	33.8	30.8 (-3.0)	27.8 (-6.0)	40.8 (+7.0)	23.8 (-10.0)	7.7	53
Madagascar	32.2	35.2 (+3.0)	32.2	41.2 (+9.0)	32.2	10.9	98
Malawi	165.9	149.9 (-16.0)	198.9 (+33.0)	212.9 (+47.0)	169.9 (+4.0)	47.4	119
Mozambique	403.7	442.7 (+39.0)	442.7 (+39.0)	442.7 (+39.0)	442.7 (+39.0)	39.0	146
Nigeria	455.6	484.6 (+29.0)	484.6 (+29.0)	484.6 (+29.0)	484.6 (+29.0)	29.5	179
Rwanda	64.1	70.1 (+6.0)	76.1 (+12.0)	83.1 (+19.0)	83.1 (+19.0)	24.8	145
Sierra Leone	32.7	35.7 (+3.0)	36.7 (+4.0)	36.7 (+4.0)	36.7 (+4.0)	4.7	125

Country	Original MOU	Best within +/-10%	Best within +/-20%	Best within +/-30%	Best, US$133.6M moved (as Higher Ceiling), +/-30%	Room to 0% cut US$M	Deaths averted by first +US$1M
Uganda	382.2	344.2 (-38.0)	306.2 (-76.0)	299.2 (-83.0)	268.2 (-114.0)	32.2	88

3. Coverage by country in the widest version (+/-30%)

Country	Δ US$M	Deaths averted 2026-30	Δ ART patients	ART coverage: baseline / Original → new	Δ people on six services	Δ bednet users
Botswana	+5.0	+768	+7,580	100.0% / 97.7% → 99.9%	+7,620	+33
Burundi	-9.0	-260	-8,618	89.4% / 81.3% → 70.8%	-159,521	-129,740
Cameroon	+16.0	+3,260	+35,248	73.0% / 66.3% → 72.9%	+635,675	+572,518
Eswatini	+10.0	+2,107	+20,227	95.2% / 84.9% → 94.3%	+20,456	+94
Ethiopia	+7.0	+1,083	+6,758	82.4% / 81.3% → 82.4%	+190,847	+137,462
Kenya	-116.0	-5,921	-164,218	91.2% / 90.0% → 78.9%	-1,906,948	-1,577,965
Lesotho	+16.0	+3,289	+32,657	96.1% / 79.5% → 92.3%	+32,990	+218
Liberia	+7.0	+464	+2,357	59.3% / 54.7% → 58.9%	+310,367	+287,071
Madagascar	+9.0	+873	+2	24.2% / 24.2% → 24.2%	+866,278	+724,181
Malawi	+47.0	+9,491	+100,133	89.1% / 79.0% → 89.0%	+773,576	+569,460
Mozambique	+39.0	+7,310	+73,174	83.7% / 80.8% → 83.7%	+398,841	+257,733
Nigeria	+29.0	+5,698	+59,319	78.1% / 75.0% → 78.0%	+1,342,539	+1,179,892
Rwanda	+19.0	+2,749	+24,710	94.3% / 80.5% → 91.0%	+629,371	+515,147
Sierra Leone	+4.0	+614	+3,410	83.9% / 79.6% → 83.3%	+332,563	+289,087
Uganda	-83.0	-4,145	-109,378	85.2% / 82.4% → 75.4%	-1,355,346	-947,967